

Sahil Khan

Software Developer | Backend & Applied AI

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2023 B.Tech Computer Science graduate focused on backend development and applied AI. Merged contributor to Turborepo.

PROJECTS

SIS Migration Engine | [GitHub](#)

TypeScript, Node.js, Express, GraphQL, PostgreSQL, Knex, BullMQ, Redis

- Built a staged CSV import workflow with Zod schemas, student-to-class reference checks, duplicate detection, and a GraphQL review flow for edits and conflict resolution.
- Queued approved records with BullMQ and Redis; used per-row PostgreSQL transactions to isolate failures and record failed rows while continuing the batch.

AmazonHelp Support Agent | [GitHub](#)

Python, Gemini API, sentence-transformers, scikit-learn, pandas

- Built a single-turn pipeline with few-shot intent classification, semantic retrieval of historical customer/reply pairs, TF-IDF fallback, and grounded reply generation.
- Added deterministic support links and escalation rules, labelled classifier datasets, an LLM rubric, blind human-scoring tools, and weighted-kappa agreement analysis.

CodebaseToSpec | [GitHub](#)

Python, AST, MongoDB, Pydantic, Gemini API

- Indexed Python files, symbols, and imports in MongoDB using AST parsing; built a bounded ReAct tool loop that produces structured plans and records tool-call traces.
- Verified cited files and indexed symbols, returning reference errors for up to three planning attempts; separated rate-limit retries from logical re-planning.

ADDITIONAL PROJECTS

[ResumeIQ](#) - React, TypeScript, Zustand, Puter.js. Uploaded resumes and job context for AI feedback; displayed structured scores and suggestions.

[DocsSphere](#) - Next.js, Clerk, Liveblocks. Integrated collaborative document editing, presence, and room-level view/edit permissions.

[CarePulse](#) - Next.js, Appwrite, Zod. Built patient registration and appointment scheduling/cancellation flows with SMS notifications.

[StoreIt](#) - Next.js, Appwrite. Implemented file upload, search, rename, sharing, and deletion; cleaned up stored files if metadata creation failed.

OPEN-SOURCE CONTRIBUTIONS

Turborepo - Rust CLI fix (merged October 2026) | [PR #14366](#)

Fixed empty task graphs in affected and tag-filtered package listings; added regression tests for task-input changes, dependency filters, and tag selectors.

Turborepo - example fix (merged June 2025) | [PR #10579](#)

Restored shared-component styling in the with-tailwind example by adding the missing Tailwind import.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, SQL

Backend & data: Node.js, Express, GraphQL, PostgreSQL, MongoDB, Redis, BullMQ, Knex, Zod

Applied AI: Gemini API, retrieval-augmented generation, sentence-transformers, TF-IDF, Pydantic

Web & tools: React, Next.js, Git, Docker

EDUCATION

Bachelor of Technology, Computer Science and Engineering | 2019 - 2023

Dr. A.P.J. Abdul Kalam Technical University